Performance Analysis of Large Language Models for Medical Text Summarization

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Abstract

Due to the rapid expansion of medical literature, keeping pace with the latest research and clinical guidelines has become more challenging for healthcare professionals. To overcome this challenge, effective text summarization is crucial for improving access to knowledge, enhancing clinical decision-making, and ultimately benefiting patient outcomes. In this study, a medical text summarization system that employs large language models (LLMs) was fine-tuned and evaluated with the objective of generating precise, logical, and brief summaries of medical literature, emphasizing clinical relevance and ease of understanding. We plan to evaluate the performance of GPT3, GPT4 and the fine-tuned T5, BART and Pegasus model trained on the standard PubMed dataset using standard evaluation metrics.

1 Introduction

There has been a constant increase in the growth of medical literature and clinical guidelines in the recent years, which has made it difficult for healthcare practitioners and researchers to stay updated with latest innovation and research in the field. Due to a lot of information, it causes delays in research and hinders decision making which can impact patient care. There has been a breakthrough in terms of advancements in Natural Language Processing (NLP) and Large Language Models (LLMs). Now, we have LLMs whose architecture have been optimized specifically for summarization tasks, and these include models like BART, Pegasus and T5. Although, these models are very effective in summarization, still their performance in medical literature needs to be thoroughly investigated and also how they compete with SoTA generative models like GPT-3 and GPT-4. The primary objective is to fine tune BART, Pegasus and T5 models on the PubMed dataset and then evaluate their performance based on popular metrics including ROUGE, METOER, Bert Score, BLEU and SacreBLEU. The models will also be compared against GPT-3 and GPT-4 models on evaluation metrics. We aim to achieve a fine-tuned model which can generate concise, logical, and accurate summaries for medical texts and presents comparable results with GPT models.

2 Related Works

In recent years, there has been an increasing interest in summarizing medical texts, due to the overwhelming volume of information and the complexity of medical language. One of the earliest approaches to text summarization involved using rule-based methods [13]. To address the limitations of rule-based methods, researchers turned to machine learning techniques. A recent study used a Naive Bayes model to summarize legal texts [1] with a ROUGE-1 score of 0.385.

Another approach to text summarization involved using deep learning models, such as RNNs and CNNs. In comparison, for summarizing legal texts, the LSTM + Glove model is used to summarize and it achieved high performance, with a ROUGE-1 score of 0.436 [1]. More recently, large language models (LLMs) gave accurate results in downstream tasks of NLP. Three notable LLMs that have shown impressive performance in text summarization are T5, Pegasus, and BART [12, 14, 6]. T5

was created by a team of researchers at Google in 2019 and is designed for text-to-text transfer learning[12]. Pegasus is a text summarization model also developed by researchers at Google in 2020[14]. BART was developed by researchers at Facebook AI Research in 2020 and is a denoising sequence-to-sequence pre-training model for language-related tasks[6]. Many studies have compared these three models, and with popular datasets like XSum, Pegasus and T5 mostly stand out as the top-2 models when evaluated on the ROUGE metric [11]. In the last few years, the sucess obtained via zero-hot and few-shot prompting with GPT models is phenomenal, and the summaries generated are overwhelmingly prefered by humans as they don't suffer from issues like poor factuality. [3] But, the zero shot prompting by text-davinci-002 when compared with finetuned models like BART and T5 scored less on rougeL metric. However, with release of GPT-3 and GPT-4, using LLMs shows great potential as a research topic to explore for text summarization.

3 Methodology

In this empirical study, we aim to compare and analyze the performance of fine-tuned three state-of-the-art (SOTA) models for abstractive text summarization: BART, Pegasus, and T5 trained on PubMed dataset. Furthermore, we evaluate the summaries generated by GPT3 & GPT-4 with the above mentioned fine-tuned models. In this section, we shall discuss the model architecture and the evaluation metrics used in this experimentation.

3.1 Fine-tuning LLMs for Summarization

3.1.1 T5

The T5 (Text-to-Text Transfer Transformer) architecture is a potent and adaptable natural language processing model created for numerous needs. In both the encoder and decoder blocks, it uses stacked multi-head self-attention and position-wise feed-forward layers, with residual connections and layer normalization applied at each step. By adding sinusoidal positional encoding to the input embeddings before processing, T5 adds positional information.[12] The self-attention mechanism, which uses multiple heads of attention, enables the model to pay attention to several input sequence elements at once and learn various contextual relationships. T5 uses a denoising autoencoder setup with a masked language modelling aim during the pre-training phase, masking input tokens at random and training the model to predict them from unmasked tokens[12]. T5 leverages the Transformer model's power while incorporating innovative approaches, such as text-to-text formatting and task-specific prefixes, making it an effective and adaptable model for numerous natural language processing tasks.

3.1.2 BART

BART, also called the Bidirectional and Auto-Regressive Transformers is a model developed by AI team at Facebook. It is used for various NLP Tasks like summarization, translation, and generation. BART[6] is based on the transformer architecture and the training steps combines the BERT's bidirectional context understanding with GPT's autoregressive generation capabilities. During preprocessing, first some of the tokens of input are masked / deleted to create a corrupted version from original sequence. The second step is the denoising part where BART learns to reconstruct the original sentence from the corrupted version. During the reconstruction, the models is able to learn the structure and semantics of the passed input. The architecture involves the encoder-decoder where encoder does the mapping for corrupted input to a latent representation, and the decoder then generates the original sentence from the representation formed by the encoder. It's architecture makes it suitable for summarization tasks and has shown lot of state-of-art results[6] on tasks like summarization of CNN / Daily Mail and XSum dataset.

3.1.3 Pegasus

Pegasus is a large language model, designed for text summarization. It is based on transformer architecture and utilizes masked language modeling for generating text. Particularly the model focuses on self-supervised summarization and gap sentence generation. The Pegasus large model had been trained with 16 layers, 1024 Hidden States fully connected layers having 4096 as out size[14]. The model size total is 568 million parameters. Some text is masked, namely mask 1, which is used for target text generation for the decoder. Particularly, it is termed as gap sentence generation, in

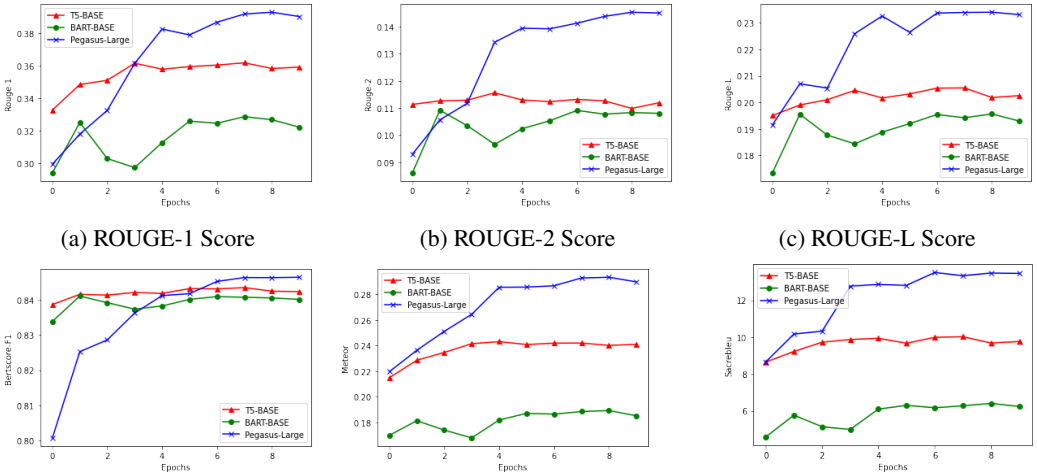
which the sentences for masking are chosen based on different strategies namely random, leading, and based on the rouge score with respect to documents. It further uses a masked language model approach and masked some text for prediction by the encoder. Different models have been trained for 12 summarization tasks.

3.2 Evaluation Metrics

The standard evaluation metrics used for summarization tasks include ROUGE scores, METEOR and Bert Scores and are used for evaluation purpose. The above mentioned metrics captures the structural and semantic similarity between the generated and reference summary. Other metrics also used for evaluation include BLEU score and SaceBLEU. The detailed discussion on these evaluation metrics has been explained in the Appendix.

4 Experimental Results

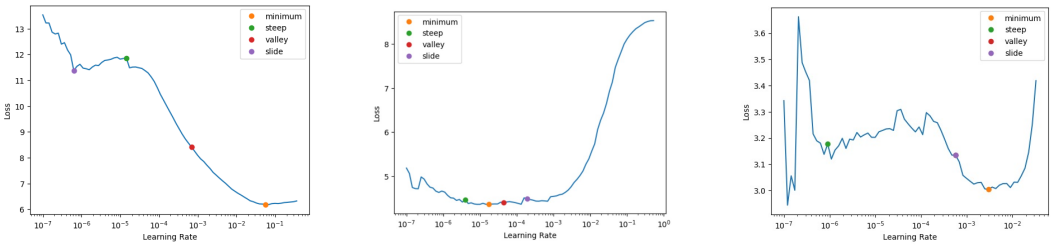
We have fine-tuned the T5-Base, BART-Base & Pegasus-Large model on 1000 samples from PubMed Summarization dataset. The hyperparameters used in training both of the models are as follows: **max_input_tok_length=1024**, **max_target_tok_length=256**, **min_summary_char_length=30**.



(a) ROUGE-1 Score (b) ROUGE-2 Score (c) ROUGE-L Score

(d) Bertscore-F1 Score (e) METEOR Score (f) SacreBLEU Score

Figure 1: Model's Training Metrics (Performance vs Epochs)



(a) T5-Base Model (b) BART-Base Model (c) Pegasus-Large Model

Figure 2: Optimal Learning Rate

Given the substantial computing resources needed for this process, it is vital to optimise the learning rate while fine-tuning large language models (LLMs). Instead of depending on trial and error, visualising the losses at each iteration with the learning rate offers a trajectory where the loss first declines and then starts to grow . The point just before the lowest loss value, which is seen as a valley in the plots, is where the most ideal learning rate may be found.

Table 1: Model performance comparison

Model	Rouge-1	Rouge-2	Rouge-L	Bertscore Precision	Bertscore Recall	BertScore F1	BLEU	SacreBLEU	METEOR
BART	0.373	0.160	0.257	0.652	0.604	0.625	0.076	8.70	0.237
T5	0.400	0.135	0.242	0.631	0.624	0.625	0.085	9.09	0.268
PEGASUS	0.440	0.174	0.250	0.637	0.661	0.646	0.118	**12.78**	0.345
GPT3	0.457	0.178	0.278	0.656	0.661	0.658	0.117	8.41	0.316
GPT4	**0.498**	**0.210**	**0.318**	**0.668**	**0.699**	**0.683**	**0.123**	10.53	**0.371**

Prompting GPT-3 & GPT-4: For GPT-3, we used the OpenAI API (text-davinci-003) to send a post request and process the output. The prompt used in experimentation is *Professionally summarize this medical article like a doctor with about [actual summary word count] words : [text]*. The hyperparameter temperature is set at 0.4 with max_tokens equals 256. For GPT-4, we manually prompted (prompt same as GPT-3) the ChatGPT-4 and noted the generated summaries (see Appendix). We evaluated all the models on 11 medical documents and the relevant metrics are reported in the table 1.

5 Discussion

As evident from the result1, the GPT-4 model outperforms all others under consideration in terms of most of the metrics. As compared to non-GPT models, it is not fine-tuned on the data. Only for the standardized version of the bleu score i.e. sacrebleu, the performance of GPT-4 is lower. It can be attributed to the paraphrasing nature of chatgpt-4 for which the tokens will not match between target and reference sentences, thus reducing Sacrebleu's score.

The Pegasus large and GPT3 models have similar performance on the metrics. Only the meteor score is relatively higher for Pegasus i.e 0.345. The reason behind that is the meteor prefers long contiguous matches with references[9]. So definitely such a metric will prefer fine-tuned model on a dataset. This aspect is further bolstered by the bert scores which are higher for GPT-3 as the metric utilizes contextual embeddings does do similarity matching by the context than token matching[15]. Size wise the Pegasus has 568 million parameters and GPT 3 has 175 billion parameters. So evidently a large zero-shot learner has been able to produce results competitive enough with a fine-tuned model.

T5 base and Bart base have 220 million and 140 million parameters, which are significantly lower than the other models under consideration. They have similar performances on rouge, bleu, and meteor scores with each other. But with significantly less computation cost they have produced results marginally poor than substantially large models. For instance, T5 and BART f1 scores are 0.625 which is bordering 0.646 for Pegasus large and 0.658 for GPT-3.

The training curves1 of the three models under consideration for different metrics plateaus after a certain number of epochs. The BART curve is oscillatory in nature initially, which can be attributed to the nature of its loss curve, which is explained later. The T5 had the highest metrics initially but Pegasaus surpassed it, after fine-tuning for a number of epochs. The In addition to the learning curve, the T5 loss decreases with increasing learning rate and stays stable. For Pegasus, there is a limited increase in the loss by increasing the learning rate after 10^{-2}. Whereas it increases significantly for BART after increasing the rate beyond the 10^{-3}. It suggests that the loss function of the BART has high curvature[2].

6 Conclusion

From the result, it is observed that GPT-4 outperforms the summarization models in their niche tasks and even without fine-tuning. Medical summarization is very technical in nature and zero-shot prediction providing competitive results reaffirms belief in the generalized nature of their learning[4]. The trends are similar to GPT-4 performance in medical competitive tests of UCMLE, thus again highlighting the complex data analysis capabilities of such LLMS[8]. As far as the comparative study of this work is considered It has been proven that very large models which are zero-shot learners give comparable or better performance than the fine-tuned models. Further, it has been observed that smaller models are marginally poor than the comparatively very large models. So these can point to future work, which can analyze the computational cost, and accuracy analysis based on the number of parameters. Even further focus on comparing the models on these attributes by fine-tuning them with larger data sets. And thus inspecting the marginal utility of spending extra computational cost.

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7 Contributions

Name	Contributions
Amandeep Singh	Fine-tuning & Inferencing of Pegasus Model, Assembling Codebase, Report Writing
Jaskaran Singh	Fine-tuning & Inferencing of BART Model, Prompting & Evaluating GPT-3, Report Writing
Tirth Patel	Fine-tuning & Inferencing of T5 Model, Prompting & Evaluating GPT-4, Report Writing

8 Appendix

8.1 Evaluation Metrics

8.1.1 ROUGE Score

A popular metric in natural language processing called ROUGE is frequently used to assess the effectiveness of text summarization software[7]. By considering the overlap of n-grams, ROUGE-N is helpful in identifying the level of similarity between the candidate summary and reference summaries. The three types of ROUGE scores we used are ROUGE-1, ROUGE-2, and ROUGE-L.

ROUGE-N: Depending on the n-gram we wanted to compare, we can compute the ROUGE-1, ROUGE-2 etc. This metric is a combination of precision and recall, which are used to determine the overlap of n-grams between the generated summary and the reference summary. This metric tries to capture the word-level similarity. ROUGE-N is calculated as:

$$\text{ROUGE-N} = \frac{\sum_{S \epsilon ReferenceSummaries} \sum_{gram_n \epsilon S} Count_{match}(gram_n)}{\sum_{S \epsilon ReferenceSummaries} \sum_{gram_n \epsilon S} Count_{match}(gram_n)}$$

ROUGE-L: Instead of focusing on N-grams like ROUGE-1/ROUGE-2, ROUGE-L uses the Longest Common Subsequence (LCS) between the generated summary and the reference summary and tries to capture the overall structure and word order within the sentences generated, making it a more comprehensive evaluation metrics.

8.1.2 BLEU

BLEU stands for Bilingual Evaluation Understudy. It is a metric used to evaluate the quality of translation by a Machine Translation System. The BLEU scores a translation on a scale of 0 to 1 where a score of 1 shows perfect translation. BLEU score can also be used for summarization tasks where we can use the summary as machine-generated text and the reference sentence is the original summary. Although, the BLEU score is not designed for summarization tasks, as it does not consider semantic similarity, and won't capture the nuances of paraphrasing and rephrasing. Hence, it only considers the overlap of similar words or word sequences, but still, it is a good metric to understand how similar the summary is when compared to the original summary. The formula for the BLEU score is given as follows:

$$\text{Bleu}_c = BP * e^{\sum_1^n w_n * log(p_n)}$$

$BLEU_c$ represents the BLEU for candidate c, which is a translation produced by a machine. BP is the brevity penalty and is a factor that accounts for the difference in the length of reference sentences and the candidate sentence.

$$BP = \begin{matrix} 1 & if\ brevity_i < 1 \\ e^{1-brevity_i} & if\ brevity_i \geq 1 \end{matrix}$$

$brevity_i = r_i/c_i$, where c_i is the length of ith candidate and r_i is the nearest length among the references. w_i is the weight assigned to each n-gram and usually has a value equal to 1/n. p_i is the precision of i-grams of the candidate sentence with respect to the reference sentences. The precision p_i is given by the formula as

$$p_i = \frac{count_i}{max(1, total_i)}$$

Where $count_i$ is the count of i-grams in candidate sentences that appear in the reference sentences. $total_i$ is the total count of i-grams in C.

8.1.3 METEOR

METEOR (Metric for Evaluation of Translation with Explicit Ordering) [5] is a metric used to evaluate quality for machine translations, summarizations, and other generative tasks. It works upon the limitations of the BLEU Score metric, where BLEU and ROUGE, as it also incorporates the linguistic features including word reordering, paraphrasing, and synonyms. The METEOR score is

given by the harmonic mean of precision and the recall score with a reordering penalty is used to consider the differences in word order.

$$METEOR = (1 - Penalty) * F_{mean}$$

Where F_{mean} is the harmonium mean of unigram precision and recall.

$$F_mean = \frac{P * R}{(\alpha * P + (1 - \alpha) * R)}$$

α here controls the importance between precision and recall. Penalty and it is given as

$$P = \gamma * (Chunkiness^{\beta})$$

Chunkiness here represents the ratio of total contiguous matches of unigrams to total unigram matches. γ and α are parameters that control the penalty strength.

8.1.4 SacreBLEU

SacreBLEU is a more consistent and standard implementation of BLEU metric which improves on various aspects like tokenization and data preparation. It works as a more standard approach to calculate the BLEU score[10], which makes sure that the results tasks are comparable across different tasks and systems. The metric calculation involves a consistent tokenization algorithm, in which things are punctuations and whitespaces are handled similarly and the preprocessing includes handling of lowercase and Unicode characters. Along with consistent techniques, it also ensures better reproducibility. This also has support for widely used test datasets like WMT, and IWSLT. WMT (Workshop on Machine Translation) and IWSLT (International Workshop on Spoken Language Translation) are two popular datasets used to evaluate the performance of machine translation systems. and makes it easier for researchers to evaluate their models and easily compare them with existing techniques. As SacreBLEU is an implementation of BLEU only, the mathematical formula is the same, the key difference is in the standardization of implementation. The value ranges from 0 to 100. Its range is different from BLEU because the BLEU score obtained is multiplied by 100 to express it in the form of a percentage. It considers a lot of factors when comparing the generated text, which includes things like stem and synonym matching, alignment, and recording along with the exact word matching.

8.1.5 BertScore

BertScore[15] is an evaluation metric that uses pre-trained contextual embedding. It matches the candidate and reference sentence, using cosine similarity. Given a reference sentence x= <$x_1, x_2..., x_n >$ and reference sentence. It is similar to similarity measures using with a difference that contextual embedding vectors are used and the vectors are normalized. Based on these vectors recall, precision and F1 scores are defined. $\hat{x} =< x\hat{}_1, \hat{x_2}, ...\hat{x_n}>$

$$R_{BERT} = \frac{1}{|x|} \sum_{x_i \epsilon x} \underset{x\hat{}_j \epsilon x\hat{}}{max} x_i^T x\hat{}_j$$

$$P_{BERT} = \frac{1}{|x\hat{}|} \sum_{x\hat{}_j \epsilon x\hat{}} \underset{x_i \epsilon x}{max} x_i^T x\hat{}_j$$

$$F_{BERT} = 2 * \frac{P_{BERT} * R_{BERT}}{P_{BERT} + R_{BERT}}$$

Further, it has been found that rare words give more information on sentence similarity. So idf with respect to sentences is defined. Given M reference sentences,$\{x(i)\}_1^M$, the idf score of a word is given as

$$idf(w) = -log \frac{1}{M} \sum_1^M \overline{\underline{||}}[w \epsilon x^i]$$

where $\overline{\underline{||}}$ is the indicator function and recall considering this idf is given as

$$R_{BERT} = \frac{\sum_{x_i \epsilon x} idf(x_i) \underset{x\hat{}_j \epsilon x\hat{}}{max} x_i^T x\hat{}_j}{\sum_{x_i \epsilon x} idf(x_i)}$$

8.2 GPT Prompts

```
prompt_prefix = "Professionally summarize this medical article like a doctor with about"
summary_wc = min(summary_wc - (summary_wc % 10), 250)

try:

  start_time = time.time()
  response = openai.Completion.create(
  model=model,
  prompt=f"{prompt_prefix} {summary_wc} to {summary_wc+50} words :\n{text}",
  temperature=0.4,
  max_tokens=toks
  )
  end_time = time.time()
  return response['choices'][0]['text'], round(end_time - start_time, 6), response['usage']['total_tokens']

except:
  pass
```

Figure 3: GPT-3 Prompt & API

Figure 4: GPT-4 Prompt

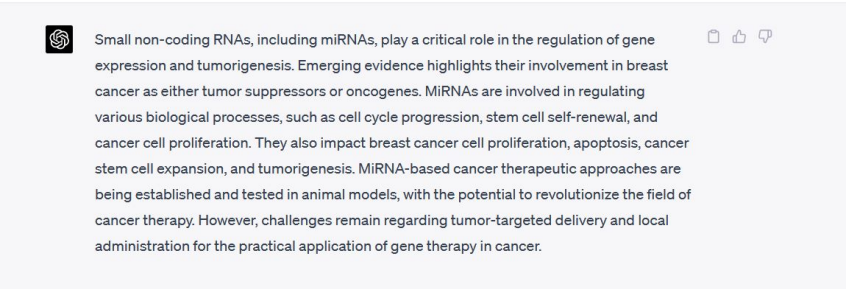

Figure 5: GPT-4 Summarization

GitHub Link: https://github.com/tiru-patel/Medical_Text_Summarization_using_LLMs